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Topic 1 — Working as a Physicist

This topic covers the fundamental tools and skills needed throughout A-Level Physics: SI units, prefixes, significant figures, estimation, experimental design, uncertainties, and graphical analysis. These skills are assessed in every paper.

1.1 SI Base Units

SI units (Systeme International d'Unites) are the internationally agreed system of measurement used in physics. All physical quantities can be expressed using combinations of seven base units.

Quantity	Unit	Symbol
Mass	kilogram	kg
Length	metre	m
Time	second	s
Electric current	ampere	A
Temperature	kelvin	K
Amount of substance	mole	mol
Luminous intensity	candela	cd

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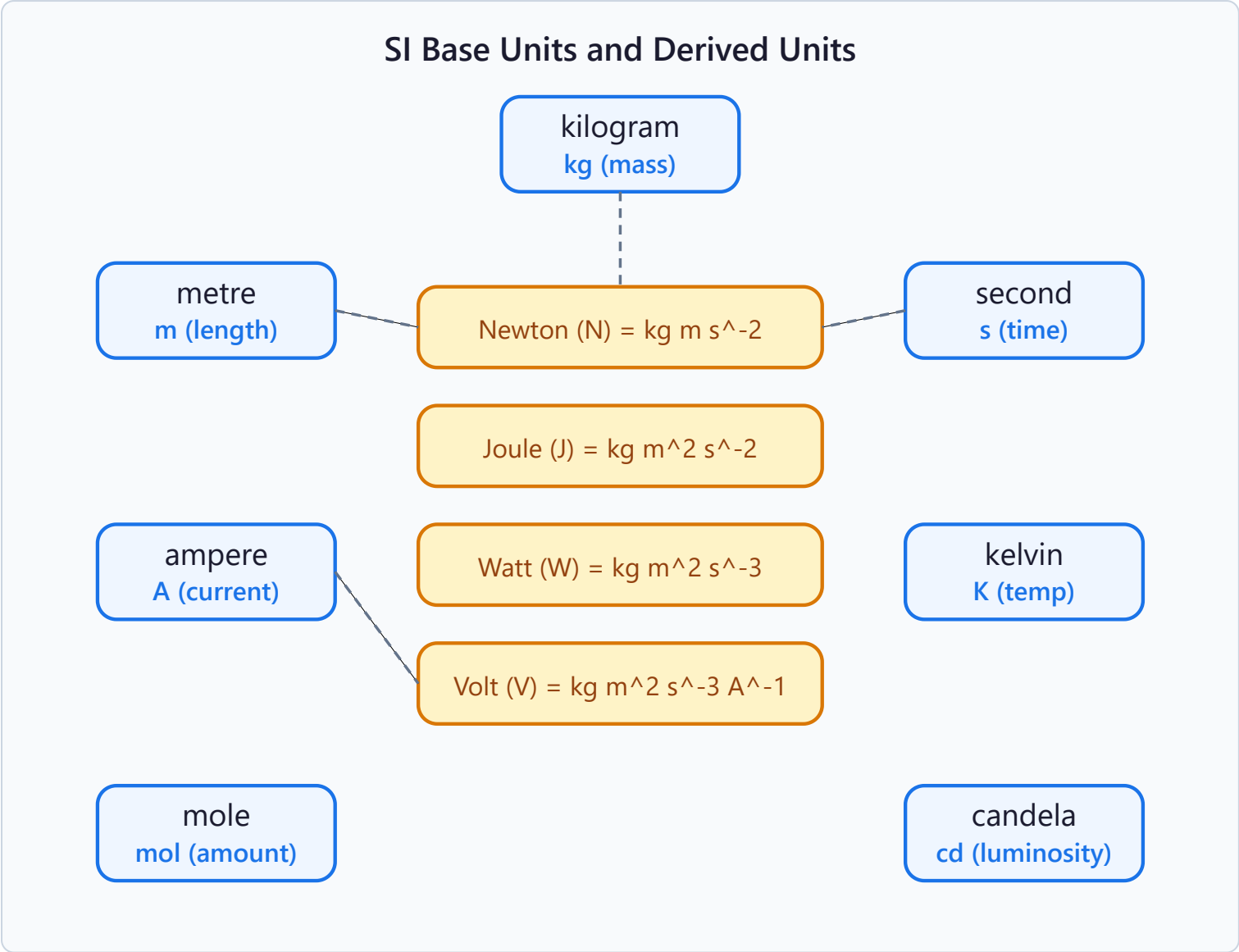

Derived Units

Derived units are combinations of base units formed by multiplying or dividing base units together. Every equation in physics must be **homogeneous** — the units on both sides must be identical.

Quantity	Derived Unit	In Base Units
Force (Newton)	N	kg m s^{-2}
Energy (Joule)	J	kg m^2 s^{-2}
Power (Watt)	W	kg m^2 s^{-3}

Pressure (Pascal)	Pa	$kg\ m^{-1}\ s^{-2}$
Charge (Coulomb)	C	A s
Potential difference (Volt)	V	$kg\ m^{2}\ s^{-3}\ A^{-1}$
Resistance (Ohm)	Ω	$kg\ m^{2}\ s^{-3}\ A^{-2}$
Frequency (Hertz)	Hz	s^{-1}
Capacitance (Farad)	F	$A^{2}\ s^{4}\ kg^{-1}\ m^{-2}$
Magnetic flux density (Tesla)	T	$kg\ s^{-2}\ A^{-1}$

Checking Homogeneity of Equations

To verify an equation is dimensionally consistent, substitute the SI base units for every term and check both sides match.

Worked Example: Verify that $E = mc^2$ is homogeneous.

Solution:
LHS: E has units of J = $kg\ m^2\ s^{-2}$
RHS: $mc^2 = kg \times (m\ s^{-1})^2 = kg \times m^2\ s^{-2} = kg\ m^2\ s^{-2}$
Both sides have units of $kg\ m^2\ s^{-2}$, so the equation is homogeneous.

Worked Example: Show that the equation $v = u + at$ is homogeneous.

Solution:
LHS: v has units of $m\ s^{-1}$
RHS: u has units of $m\ s^{-1}$; at has units of $m\ s^{-2} \times s = m\ s^{-1}$
$u + at = m\ s^{-1} + m\ s^{-1} = m\ s^{-1}$. Both sides match.

Exam Tip: To check if an equation is **homogeneous**, substitute the SI base units into both sides. If they match, the equation is dimensionally consistent. However, a homogeneous equation may still have incorrect numerical constants or be missing dimensionless factors — homogeneity is necessary but not sufficient for correctness.

1.2 SI Prefixes

Prefixes are used to represent very large or very small quantities concisely. Always convert to base SI units before substituting into equations.

Prefix	Symbol	Multiplier	Power of 10
tera	T	1,000,000,000,000	10^{12}
giga	G	1,000,000,000	10^{9}
mega	M	1,000,000	10^{6}
kilo	k	1,000	10^{3}
centi	c	0.01	10^{-2}
milli	m	0.001	10^{-3}
micro	μ	0.000001	10^{-6}
nano	n	0.000000001	10^{-9}
pico	p	0.000000000001	10^{-12}
femto	f	10^{-15}	10^{-15}

Converting Units

To convert between prefixed units, multiply or divide by the appropriate power of 10:

- 3.5 km = 3.5×10^3 m = 3500 m
- 250 nm = 250×10^{-9} m = 2.50×10^{-7} m
- 4.7 MΩ = 4.7×10^6 Ω
- 15 μF = 15×10^{-6} F = 1.5×10^{-5} F

Exam Tip: A very common error is forgetting to convert prefixed units before substituting into equations. For example, if a diameter is given as 0.50 mm, you must convert to 5.0×10^{-4} m before using it in $A = \pi r^2$. Pay particular attention to squared and cubed units: 1 cm^2 = 10^{-4} m^2 (not 10^{-2}), and 1 cm^3 = 10^{-6} m^3 (not 10^{-3}).

1.3 Significant Figures and Decimal Places

Significant figures (s.f.) are the number of meaningful digits in a value. They indicate the precision of a measurement.

Rules for Counting Significant Figures

- All non-zero digits are significant: 3456 has 4 s.f.
- Zeros between non-zero digits are significant: 3006 has 4 s.f.
- Leading zeros are NOT significant: 0.0045 has 2 s.f.
- Trailing zeros after a decimal point ARE significant: 2.50 has 3 s.f.
- Trailing zeros without a decimal point are ambiguous: 1500 could be 2, 3, or 4 s.f. (use scientific notation to be clear: 1.50×10^3 = 3 s.f.)

Using Significant Figures in Calculations

- **Multiplication / Division:** The answer should have the same number of s.f. as the value with the fewest s.f.
- **Addition / Subtraction:** The answer should have the same number of decimal places as the value with the fewest d.p.
- Final answers are typically given to 2 or 3 s.f. in A-Level Physics.

- Do not round intermediate calculations — only round the final answer.

Exam Tip: If the question gives data to 3 s.f., give your answer to 2 or 3 s.f. Never give more significant figures in your answer than in the data you were given. A common mark scheme note is "accept 2 or 3 s.f." for the final answer.

1.4 Estimation and Orders of Magnitude

Order of magnitude: The power of 10 closest to a quantity. Two values are of the same order of magnitude if they are within a factor of 10 of each other.

Common Estimations You Should Know

Quantity	Approximate Value	Order of Magnitude
Mass of an electron	9.1×10^{-31} kg	10^{-30} kg
Mass of a proton	1.7×10^{-27} kg	10^{-27} kg
Diameter of an atom	~10^{-10} m	10^{-10} m
Diameter of a nucleus	~10^{-15} m	10^{-15} m
Wavelength of visible light	400–700 nm	10^{-7} m
Height of a person	~1.7 m	10^{0} m
Mass of a person	~70 kg	10^{2} kg
Mass of a car	~1500 kg	10^{3} kg
Radius of the Earth	6.4×10^{6} m	10^{7} m
Speed of sound in air	~340 m s^{-1}	10^{2} m s^{-1}

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Speed of light	3×10^8 m s^{-1}	10^8 m s^{-1}
Atmospheric pressure	~1×10^5 Pa	10^5 Pa

Worked Example: Estimate the number of atoms in a human body.

Solution:
Mass of a person ≈ 70 kg. The human body is mostly water, so treat as water (molar mass = 18 g/mol = 0.018 kg/mol).
Number of moles ≈ 70 / 0.018 ≈ 3900 mol
Number of molecules ≈ $3900 \times 6 \times 10^{23} \approx 2.3 \times 10^{27}$ molecules
Each water molecule has 3 atoms, so number of atoms ≈ $3 \times 2.3 \times 10^{27} \approx$ **7×10^{27}** atoms
Order of magnitude: **10^{28}**

Exam Tip: For estimation questions, show your reasoning clearly. State your assumptions (e.g. "assume the body is mostly water"). An answer within an order of magnitude of the "correct" answer typically earns full marks. The process is more important than the exact number.

1.5 Planning Experiments and Variables

Independent variable (IV): The variable you deliberately change during the experiment.
Dependent variable (DV): The variable you measure as a result of changing the IV.
Control variables: All other variables that must be kept constant to ensure a fair test.

Planning an Experiment

A complete experimental plan should include:

- Clearly stated IV, DV, and all significant control variables with how each is maintained
- List of apparatus with specifications (e.g. "micrometer screw gauge, resolution 0.01 mm")
- A step-by-step method that could be replicated by another physicist
- The range and intervals of the IV (at least 5–6 values, spread across a suitable range)

- Number of repeat readings (at least 3) and calculation of mean
- Safety precautions and risk assessment
- What to plot and how to analyse the graph (e.g. "plot R vs L; gradient = ρ/A")

Types of Error

Type	Description	Effect on Results	How to Reduce
Random error	Unpredictable variation between measurements, causing scatter around the true value	Results scattered both above and below the true value	Take repeat readings and calculate the mean; use more precise instruments
Systematic error	A consistent error in the same direction every time (e.g. a zero error, miscalibrated instrument)	All results shifted in the same direction from the true value	Recalibrate instruments; check for zero errors; use a different method
Zero error	A specific type of systematic error where an instrument does not read zero when it should	All readings are offset by a constant amount	Check and record the zero error; subtract it from all readings

Accuracy vs Precision

Accuracy: How close a measurement is to the true value. Affected by systematic errors.
Precision: How close repeat measurements are to each other. Affected by random errors.
Repeatability: How close results are when the same person repeats the experiment with the same equipment.
Reproducibility: How close results are when different people repeat the experiment, or when using different equipment or methods.

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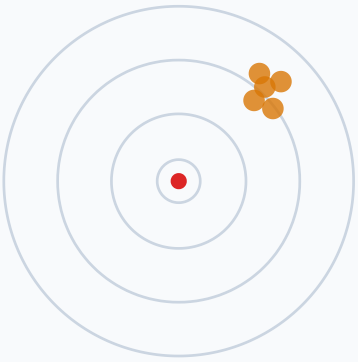
Accuracy vs Precision (Target Board Analogy)

Precise, NOT Accurate



Clustered (low scatter) but offset
Systematic error present

Accurate, NOT Precise

Scattered (high spread) but centred
Random error dominates

1.6 Uncertainties

Uncertainty: A measure of the range within which the true value is likely to lie. It quantifies the confidence in a measurement.

Absolute Uncertainty

The absolute uncertainty is expressed in the same units as the measurement.

- **For a single reading:** Uncertainty = ± resolution of the instrument (or ± half the smallest division for analogue instruments).
- **For repeated readings:** Uncertainty = (max value − min value) / 2. Use whichever is larger (resolution or half-range).

Percentage Uncertainty

$$\text{Percentage uncertainty} = (\text{absolute uncertainty} / \text{measured value}) \times 100\%$$

Combining Uncertainties

Operation	Rule	Example
Adding or subtracting: $y = a \pm b$	Add the **absolute** uncertainties: $\Delta y = \Delta a + \Delta b$	If $a = 5.0 \pm 0.1$ and $b = 3.0 \pm 0.2$, then $a - b = 2.0 \pm 0.3$
Multiplying or dividing: $y = a \times b$ or $y = a/b$	Add the **percentage** uncertainties: $\%\Delta y = \%\Delta a + \%\Delta b$	If $\%\Delta a = 2\%$ and $\%\Delta b = 3\%$, then $\%\Delta y = 5\%$
Raising to a power: $y = a^n$	Multiply the percentage uncertainty by n: $\%\Delta y = n \times \%\Delta a$	If a has 2% uncertainty, then a^2 has 4% uncertainty and $\sqrt{a}$ has 1%

Worked Example: A wire has diameter $d = 0.52 \pm 0.01$ mm. Calculate the percentage uncertainty in the cross-sectional area $A = \pi d^2/4$.

Solution:
% uncertainty in d = (0.01/0.52) × 100% = 1.92%
Since A depends on d^2, the % uncertainty in A = 2 × 1.92% = **3.85%** (approximately 4%)
This is why diameter is often the largest source of uncertainty in resistivity experiments.

Exam Tip: When a quantity is squared (e.g. v^2 or d^2), the percentage uncertainty is **doubled**. When a quantity is square-rooted, the percentage uncertainty is **halved**. This is one of the most commonly tested uncertainty rules.

1.7 Graphs

Drawing Graphs

- Use at least half the available graph paper for your plotted points (use appropriate scales).
- Label both axes with the quantity and its unit, separated by a slash or in brackets: e.g. "velocity / m s^{-1}" or "velocity (m s^{-1})".
- Choose sensible scales — use easy divisions (1, 2, 5, or 10 per large square). Avoid awkward scales like 3 or 7 per square.

- Plot points accurately with small, precise crosses (not dots or circles).
- Draw a line of best fit — this can be a straight line or a smooth curve. It does not need to pass through every point, but should have roughly equal numbers of points on each side.
- Do not force the line through the origin unless the theory predicts it should pass through (0, 0).

Gradient

$$\text{gradient} = \Delta y \,/\, \Delta x = (y_2 - y_1) \,/\, (x_2 - x_1)$$

Use a large triangle (at least half the line length) for accuracy. Choose two points **on the line of best fit**, not raw data points. The gradient often represents a physical quantity (e.g. gradient of v-t graph = acceleration; gradient of R-L graph = ρ/A).

Y-Intercept

The y-intercept is the value of y when x = 0. Read it directly from the graph if the y-axis starts at zero. If the axis does not start at zero, calculate it using y = mx + c with a point on the line and the gradient m. The y-intercept often represents a physical constant (e.g. in V = ε − Ir, the y-intercept of V vs I is ε).

Error Bars and Lines of Best/Worst Fit

- **Error bars** extend above and below each point by the size of the absolute uncertainty.
- Draw a **best fit line** that passes as close to as many points as possible.
- Draw **worst fit lines** — the steepest and shallowest lines that still pass through all error bars.
- Uncertainty in gradient = (steepest gradient − shallowest gradient) / 2.
- Uncertainty in y-intercept = (steepest intercept − shallowest intercept) / 2.

Linearising Equations

Non-linear relationships can be plotted as straight lines by choosing appropriate variables. Compare the equation to y = mx + c:

Relationship	Linearised Form	Plot	Gradient	Intercept
$y = kx^2$	$y = k(x^2)$	y vs x^2	k	0

$y = k/x$	$y = k(1/x)$	y vs $1/x$	k	0
$y = ax^n$	$\log y = n \log x + \log a$	$\log y$ vs $\log x$	n	$\log a$
$y = Ae^{kx}$	$\ln y = kx + \ln A$	$\ln y$ vs x	k	$\ln A$
$T^2 = (4\pi^2/g)L$	$T^2 = (4\pi^2/g)L$	T^2 vs L	$4\pi^2/g$	0

Worked Example: An experiment gives data that is expected to follow $I = I_0e^{-\lambda t}$. What should be plotted to produce a straight line, and how are I_0 and λ determined?

Solution:
Take natural log of both sides: $\ln(I) = \ln(I_0) - \lambda t$
This has the form $y = mx + c$, so plot **ln(I) vs t**.
Gradient = −λ (so λ = magnitude of gradient).
Y-intercept = $\ln(I_0)$, so $I_0 = e^{\text{intercept}}$.

Exam Tip: Always use data points that lie ON your line of best fit when calculating the gradient, not raw data points from the table. The two points should be far apart for maximum accuracy. State the coordinates of both points and show your working.